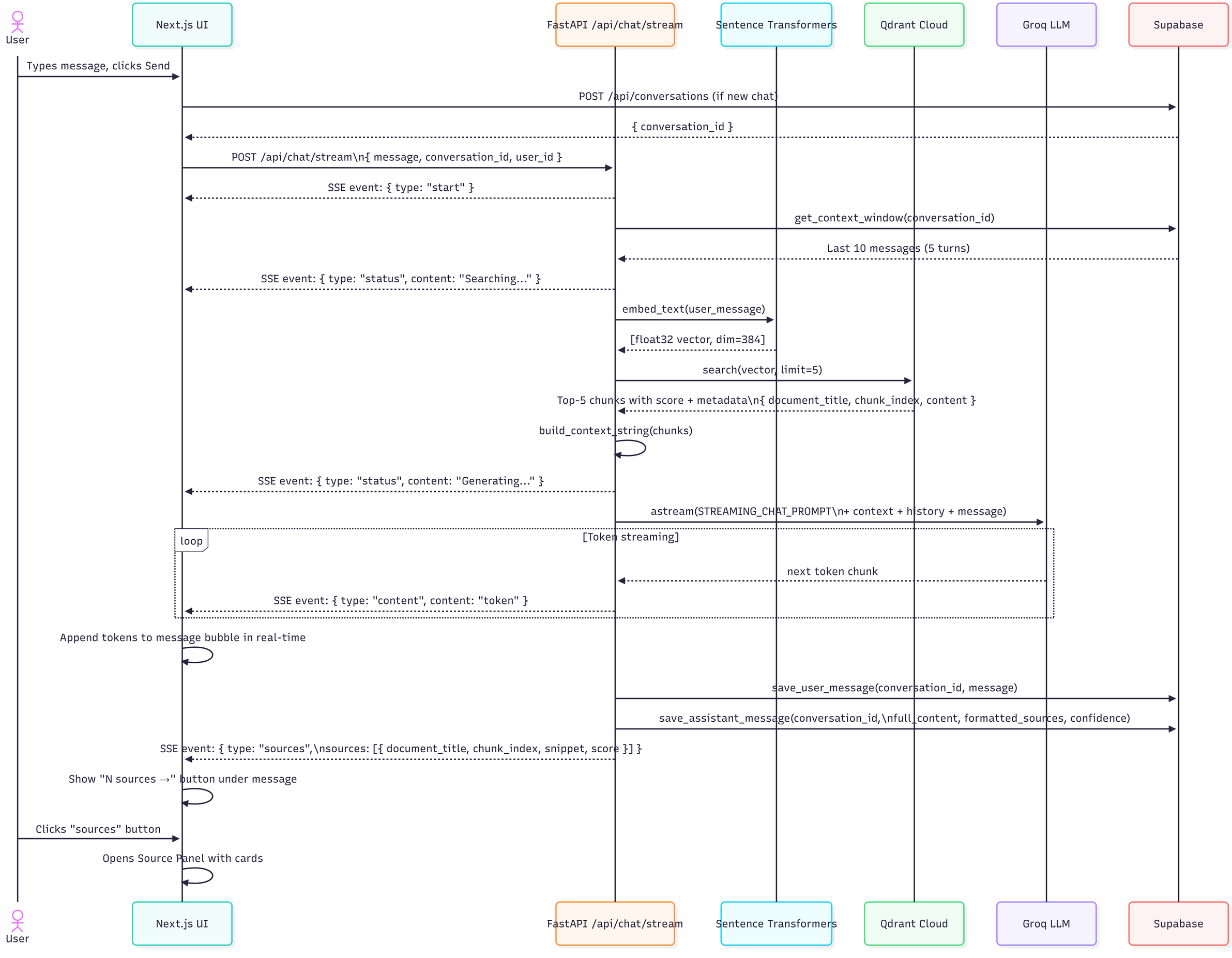
User
Next.js UI
FastAPI /api/chat/stream
Sentence Transformers
Qdrant Cloud
Groq LLM
Supabase
Types message, clicks Send
POST /api/conversations (if new chat)
{ conversation_id }
POST /api/chat/stream\n{ message, conversation_id, user_id }
SSE event: { type: "start" }
get_context_window(conversation_id)
Last 10 messages (5 turns)
SSE event: { type: "status", content: "Searching..." }
embed_text(user_message)
[float32 vector, dim=384]
search(vector, limit=5)
Top-5 chunks with score + metadata\n{ document_title, chunk_index, content }
build_context_string(chunks)
SSE event: { type: "status", content: "Generating..." }
astream(STREAMING_CHAT_PROMPT\n+ context + history + message)
loop
[Token streaming]
next token chunk
SSE event: { type: "content", content: "token" }
Append tokens to message bubble in real-time
save_user_message(conversation_id, message)
save_assistant_message(conversation_id,\nfull_content, formatted_sources, confidence)
SSE event: { type: "sources",\nsources: [{ document_title, chunk_index, snippet, score }] }
Show "N sources →" button under message
Clicks "sources" button
Opens Source Panel with cards